

Quartapelle's Projection Condition

Tags: #Fluid-Mechanics #Numerical-Method #Differential_Equations

Motivation

The proper conditioning for the vorticity is of an integral (nonlocal) character and consists in fixing *a projection of the vorticity field on the linear manifold of the harmonic vector fields*. This circumstance indicates that, once the problem has been stated in terms of boundary values for the stream function and its normal derivative, it will not admit an exact reformulation in terms of vorticity boundary values, except when the vorticity on the boundary is specified from the very outset, as, for instance, on planes of symmetry. As a practical consequence, *in numerical calculations it is impossible to devise a boundary vorticity formula which will prove effective in the generality of cases*. On the other hand, the integral projection conditions allow a splitting of the biharmonic equation into two secondorder equations to be solved in sequence. A similar, yet incomplete, splitting is also possible for the general case of nonlinear equations in two and three dimensions.

Question

What is the condition which allows us to split the problem

ψ : a partial differential equation of order 4

$$\psi|_S = a,$$

$$\frac{\partial \psi}{\partial n}|_S = b$$

into a system of equations such that

$$\begin{cases} \nabla^2 \psi &= \zeta \\ \nabla^2 \zeta &= f \end{cases}$$

where f is a function (*the source term of the Poisson equation*) defined in the domain.

Theorem: The Projection Condition

A function ζ in V is such that $\zeta = \nabla^2 \psi$, with $\psi|_S = a$ and $\frac{\partial \psi}{\partial n}|_S = b$, if and only if

$$\int_V \zeta \eta dV = \oint_{\partial V} (b\eta - a \frac{\partial \eta}{\partial n}) ds$$

for any function harmonic η in V .

The proof is straightforward since it only applies the second Green's Identity, which is

$$\begin{aligned} \int \int_V (u \nabla^2 v - v \nabla^2 u) dx dy &= \oint_{\partial V} (u \nabla v \cdot \mathbf{n} - v \nabla u \cdot \mathbf{n}) ds \\ &= \oint_{\partial V} (u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}) ds \end{aligned}$$

Applying this formula to the integral and assuming that η is harmonic in V , we have

$$\begin{aligned} - \int_V \zeta \eta dV &= - \int_V (\nabla^2 \psi) \eta dV \\ &= \int_V (\nabla^2 \eta) \psi dV - \oint_{\partial V} (\nabla^2 \psi) \eta dV \\ &= \oint_{\partial V} (\psi \frac{\partial \eta}{\partial n} - \eta \frac{\partial \psi}{\partial n}) ds \end{aligned}$$

Thus,

$$\int_V \zeta \eta dV = \oint_{\partial V} (b\eta - a \frac{\partial \eta}{\partial n}) ds$$

Now, we need to conversely prove that if ζ satisfies this relation, then

$$b = \frac{\partial \psi}{\partial n}|_S$$

Let ν be a function such that

$$\int_V \nu \eta dV = \oint_{\partial V} (b\eta - a \frac{\partial \eta}{\partial n}) ds$$

and let ψ be the unique solution of the Poisson equation

$$\nabla^2 \psi = \nu$$

with the Dirichlet boundary condition $\psi|_S = a$. By the Green's identity,

$$\begin{aligned} \int_V \nu \eta dV &= \int_V (\nabla^2 \psi) \eta dV \\ &= \int_V \psi \nabla^2 \eta dV + \oint_{\partial V} (\frac{\partial \psi}{\partial n} \eta - \psi \frac{\partial \eta}{\partial n}) ds \\ &= \oint_{\partial V} (\frac{\partial \psi}{\partial n} \eta - a \frac{\partial \eta}{\partial n}) ds \end{aligned}$$

Thus,

$$\oint_{\partial V} b\eta \, ds = \oint_{\partial V} \frac{\partial \psi}{\partial n} \eta \, ds$$

Remarks

- The left hand side of the condition is the scalar product of ζ and η in the Hilbert space L^2 of square integrable functions. When $a = b = 0$, it is orthogonal in the L^2 sense to the *manifold of the harmonic functions*. In the general case of non-homogeneous boundary conditions for ψ , the projection of ζ on this manifold is determined by a and b .
- It is noted that the projection conditions lead to the complete decoupling of the two Poisson equations which result from the bi-harmonic equation. Therefore, ζ can be calculated directly and no iterative method is required, as in the case of the coupled equation approach to the bi-harmonic problem.

Two-Dimensional Equations

For two-dimensional equations, the problem can be stated as

$$\begin{aligned} \frac{1}{Re} \nabla^2 \zeta &= \frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla) \zeta, \\ \int_V \zeta \eta \, dV &= \oint (b\eta - a \frac{\partial \eta}{\partial n}) \, ds, \\ \forall \eta : \nabla^2 \eta &= 0 \\ \nabla^2 \psi &= \zeta, \\ \psi|_S &= a, \\ \frac{\partial \psi}{\partial n}|_S &= b \end{aligned}$$

The projection of the vorticity field ζ on the linear manifold of the harmonic functions is determined by the instantaneous values of the tangential and normal velocity along all the boundary. *It is clear that these integral conditions cannot be reduced to conditions of a local type specifying the boundary values of the vorticity.*

The corresponding numerical scheme is called Influence Matrix Method. Related note: [[Numerical Application of Quartapelle's Projection Condition (Time Invariant Case)]]

The 3D case

(To be continued.....)

Thought-Math/Physical Insight

- This theorem gives us a very interesting information about the vorticity field, because it tells us that the “distribution” of a vorticity field over a domain depends on the whole boundary conditions (Dirichlet and Neumann) and the boundary conditions of the test function. For the particular case of fixed boundary and no-slip condition, the projection condition will be reduced to orthogonality of the vorticity field in the domain. The related topic about the basis of harmonic space: *Harmonic Measure*
- There is another interesting observation about the projection condition. Let us recall the definition of circulation

$$\int_V \omega dV = \oint_{\partial V} \mathbf{u} \cdot d\mathbf{s}$$

where $d\mathbf{s} = \mathbf{t} ds$ and $\mathbf{t}$ is the tangent vector of the boundary. It seems that its mathematical expression corresponds to the a particular case of the projection such that $\eta \equiv 1$. Perhaps it provides us a hint: the information which we want to derive from the vorticity field depends on the projection on a certain function space. When we project the vorticity with $\eta \equiv 1$, it gives us the integral over the domain; if we project it with harmonic functions, then we will obtain its distribution over the domain, which is limited by some boundary conditions.

Question:

- Why did Quartapelle use harmonic function as test function?
- Can we use other functions to present the integral constraint ?
- Assuming that the distribution of the vorticity is unique, is it possible to set a particular projection so that we can directly obtain its boundary value?

Reference:

- Quartapelle, L., and Valz-Gris, F. (1981). Projection conditions on the vorticity in viscous incompressible flows. Int. J. Num. Meth. FI. 1, 129.
- M Napolitano, G Pascazio , L Quartapelle ,A review of vorticity conditions in the numerical solution of the $\zeta - \psi$ equations, Computers & Fluids 28 (1999) 139-185